Orbit Propagation 1

Elizabeth Klioner M.Sc. | Chair of Space Technology | Institute of Aeronautics and Astronautics (ILR)

Orbit Geometry

Keplerian Elements (1)

6 Kepler elements

a semi-major axis

i inclination

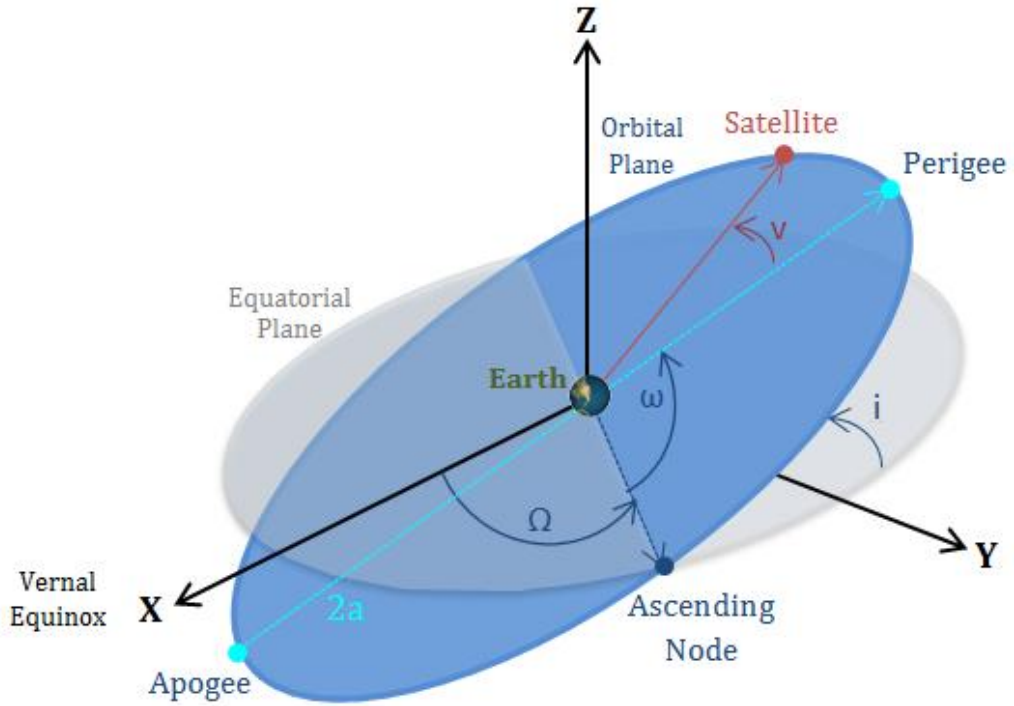
e eccentricity

Ω rigaht ascension of ascending node

ω argument of perigee

v true anomaly

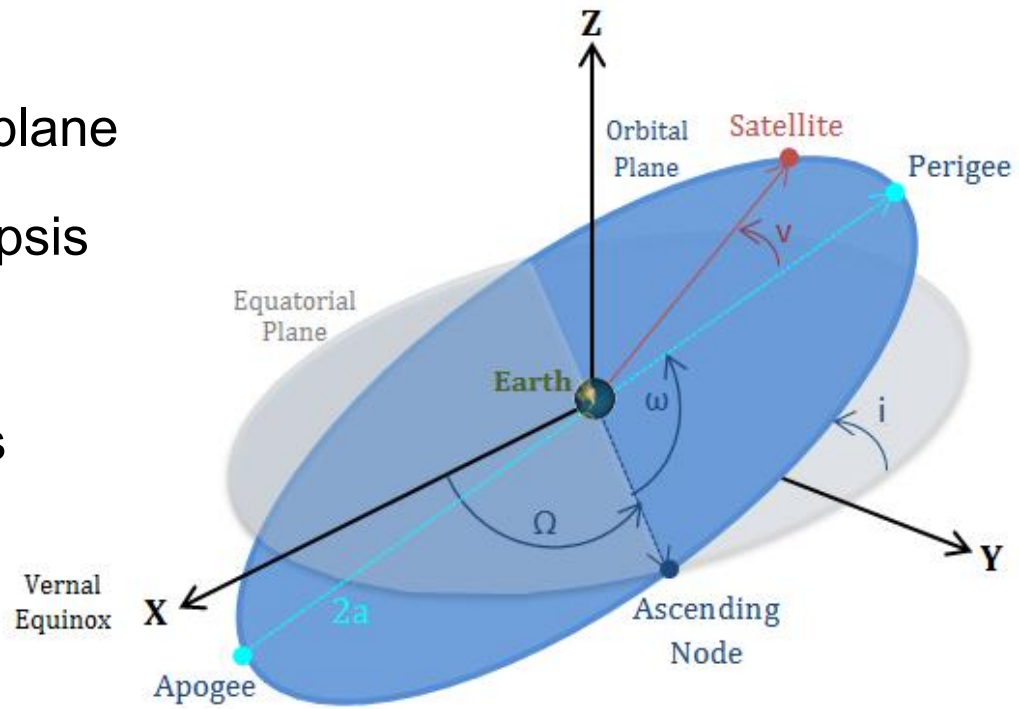
Orbit equation: $r = \frac{a(1-e^2)}{1+e\cdot\cos(v)}$



[2]

Keplerian Elements (2)

- Geometric quantities
  - Ω, i describe the orientation of the orbital plane
  - ω describes the orientation of the orbit ellipsis in the orbital plane
  - a, e describe the shape of the orbit ellipsis
  - ν describes the position of the satellite on its orbit
- Defined in an inertial frame



[2]

Orbit Shapes

Ellipse

Gravitational parameter μ (also called GM)

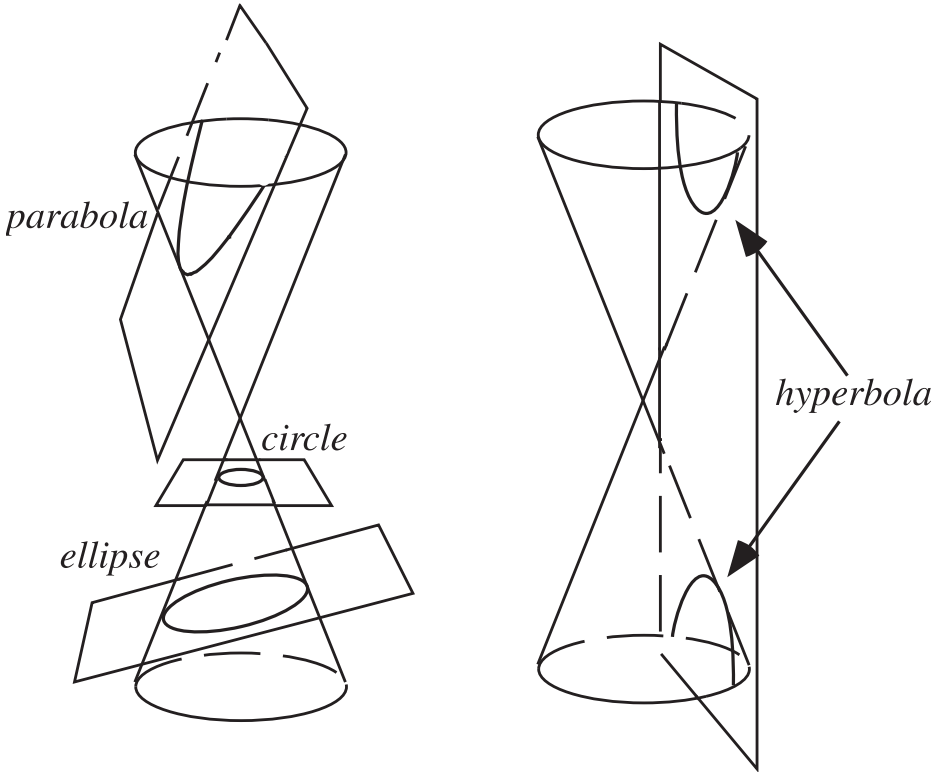
$$a > 0, 1 > e \geq 0, v = \sqrt{\mu\left(\frac{2}{r} - \frac{1}{a}\right)}$$

Parabolic

$$a \to \infty, e = 1, v = \sqrt{\mu\left(\frac{2}{r}\right)}$$ (2. cosmic velocity, i.e. escape velocity)

Hyperbolic

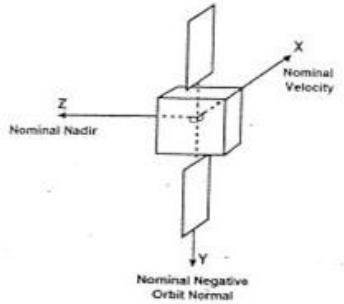
$$a < 0, e > 1, v = \sqrt{\mu\left(\frac{2}{r} - \frac{1}{a}\right)}, \text{ with } v_{HE} = \sqrt{\mu\left(\frac{1}{a}\right)}$$ (hyperbolic excess velocity)



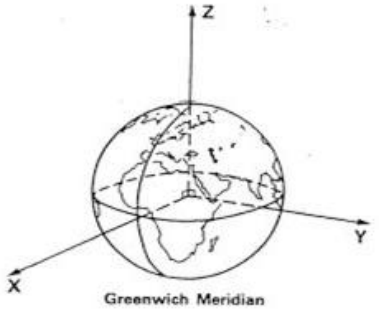
[3]

Reference System Classification

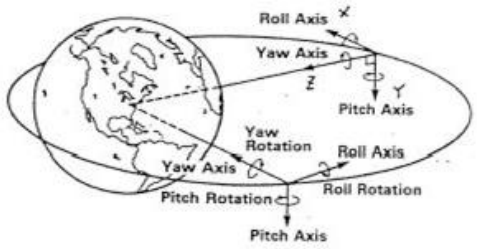
- Body Fixed
  - e.g. COS of instruments
- Path Fixed
  - radius or velocity based
- Earth Fixed
  - Fixed to the central body
- Inertially Fixed
  - Inertial frame, i.e. not accelerated



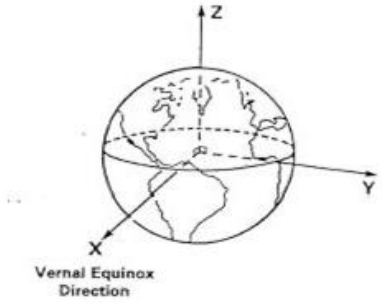
Spacecraft body Coordinates



Earth-Centered Earth-Fixed (ECEF) Coordinates



Local Vertical Local Horizontal (LVLH) Coordinates



Earth Centered Inertial (ECI) Coordinates

[4]

International Celestial Reference System (ICRS) and Frame (ICRF)

- Reference System:
  - Definition of origin and axes
  - Specification of fundamental models to convert between observation data and reference data
  - Example: ICRS (International Celestial Reference System)
- Reference Frame:
  - Practical Realization of Reference System
  - Catalogue of celestial objects defining the orientation
- ICRF is the realization of the ICRS

Motion of a Satellite

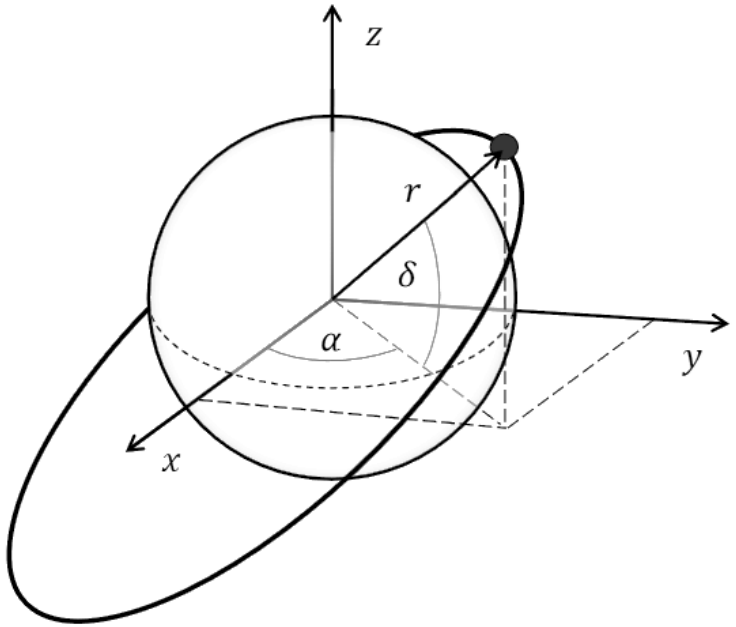
- Satellites move in their orbital plane
- Position vector $\boldsymbol{r}$

  velocity vector $\dot{\boldsymbol{r}} = \boldsymbol{v}$

  angular momentum vector $\boldsymbol{h}$
- Law of conservation of angular momentum: $\boldsymbol{h} = \boldsymbol{r} \times \dot{\boldsymbol{r}} = \boldsymbol{const.}$
- Kepler's seconds law: $\Delta A = \frac{1}{2}|\boldsymbol{h}|\Delta t = \text{const.}$
- $\boldsymbol{r}$ and $\boldsymbol{v}$ are the state of the satellite

Cartesian & Spherical Representation in Space

- The position of a satellite can be given in
  - Cartesian coordinates x, y, z
  - Spherical coordinates α, δ, r

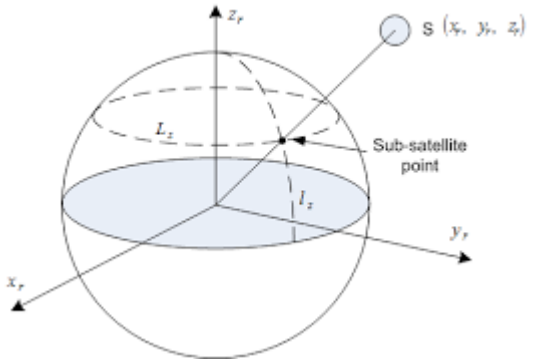


$$\boldsymbol{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = r \begin{pmatrix} cos\delta\ cos\alpha \\ cos\delta\ sin\alpha \\ sin\delta \end{pmatrix}$$

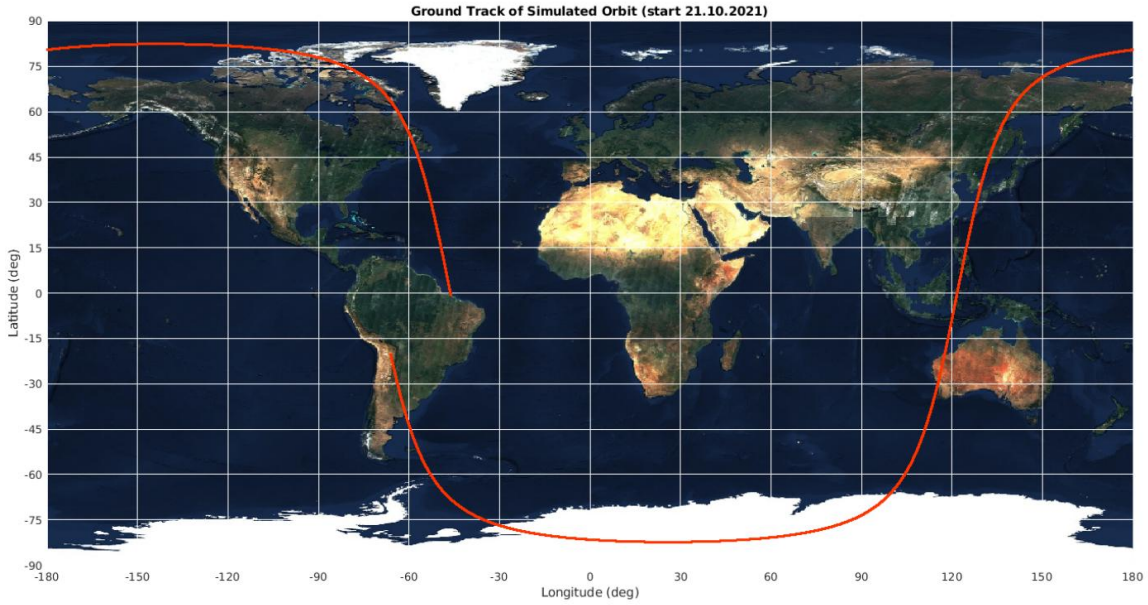
➔ $\alpha = arctan\left(\frac{x}{y}\right), \delta = arctan\left(\frac{z}{\sqrt{x^2+y^2}}\right), r = \sqrt{x^2+y^2+z^2}$

Sub-Satellite Point and Ground Track (1)

- Sub-satellite point: point on Earth's surface in nadir direction of the satellite
- Ground track: subsatellite points propagated over time



[5]



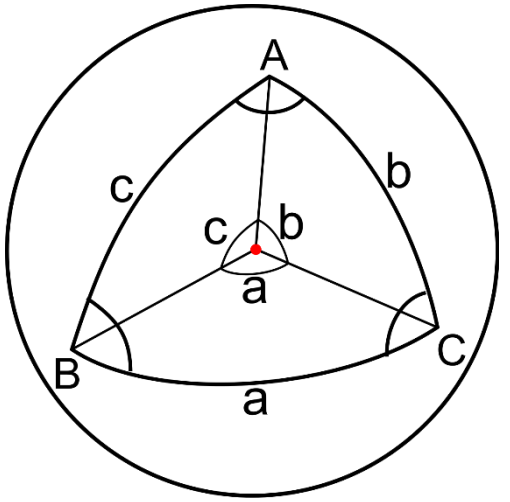
Sub-Satellite Point and Ground Track (2)

- Sub-satellite point is calculated as cartographic coordinates on the surface of the central body, e.g. Earth
  - Coordinates are angles! λ latitude, ϕ longitude
- Cartographic coordinate systems are either spherical or geodetic
  - Spherical coordinates are an approximation, Earth is not a perfect sphere
  - Geodetic coordinate systems model Earth as an ellipsoid
- Cartographic coordinate systems exists for many bodies of the Solar System
  - See https://link.springer.com/article/10.1007/s10569-010-9320-4

Recap: Spherical Geometry and Trigonometry

- Geometry: surface of a sphere in general
- Trigonometry: sides and angles of triangles on spheres
- Different geometrical rules than in Euclidian geometry (e.g. sum of all angles in a triangle ≠ 180°)
- Sides of triangles are given as angles

  ➔ cosine and sine rules
- For small areas a plane is a sufficiently accurate simplification



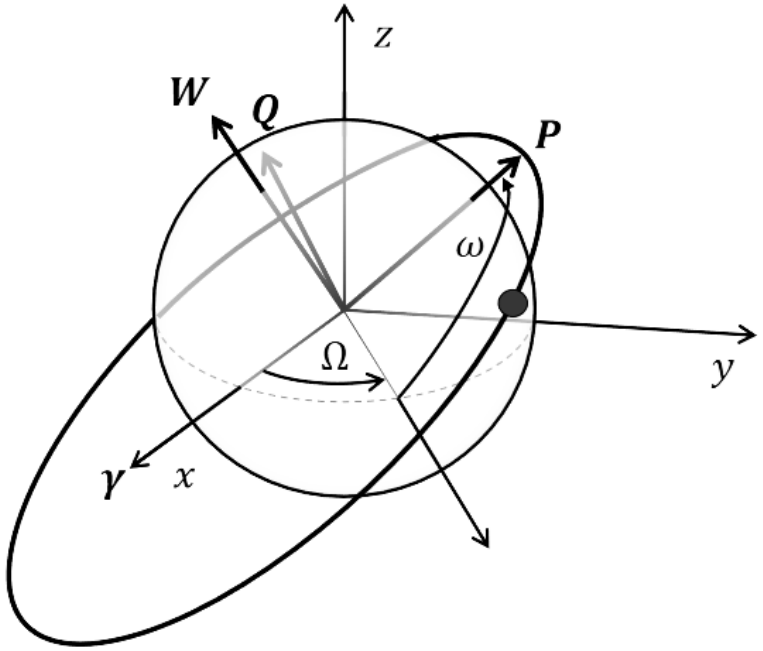
[1]

Perifocal System (PQW)

… coordinate system with orbital plane as fundamental plane

- Base vectors:
  - P: pointing towards perigee
  - W: in direction of orbital momentum vector $\boldsymbol{h}$
  - Q: completing the right hand system,

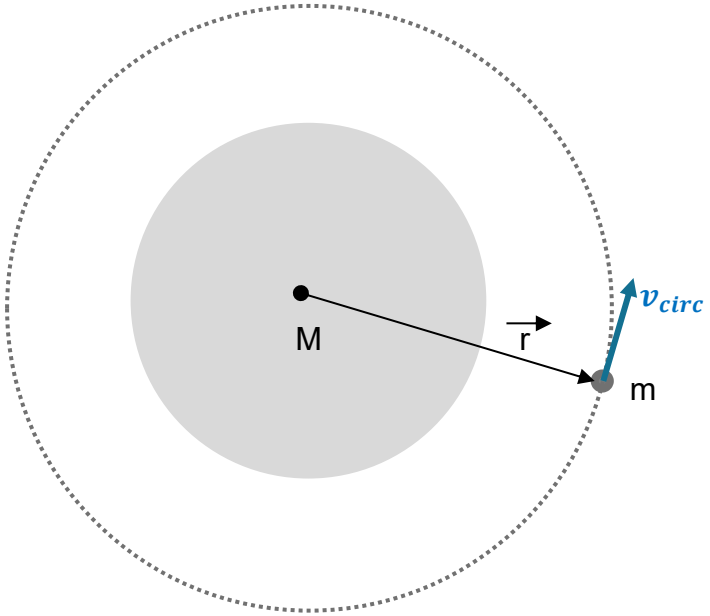
    pointing towards $\nu = 90°$



Modelling Unperturbed Orbits

Basis: (Restricted) Two-Body Problem

- Satellite with mass m moving with constant speed around body with mass M
- Gravitational force = centripetal force



$$\vec{a}_{cp} = \ddot{\vec{r}} = -\frac{v^2}{r^2}\vec{r}$$

$$\vec{F}_{cp} = \vec{F}_G$$

$$\vec{F}_{cp} = m \cdot \vec{a}_{cp}$$

$$\ddot{\vec{r}} \cdot m = \frac{GM\,m}{r^3}\vec{r}$$ (Newton's law of gravitation)

$$\ddot{\vec{r}} = \frac{\boldsymbol{GM}}{\boldsymbol{r^3}}\vec{\boldsymbol{r}}$$ ➔ $v_{circ} = \sqrt{GM/r}$

What happens when “propagating an orbit”?

$$\ddot{\vec{r}} = \frac{GM}{r^3}\vec{r}$$

$$\ddot{\vec{r}} = \frac{d\dot{\vec{r}}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

➔ You want to solve for $\vec{r}(t)$ to get your position at time t

➔ Laws of nature will give you forces, that you can express as accelerations

➔ With unperturbed orbits the only force you consider is the gravitational force

$$\vec{a}_{grav.} = \frac{d\dot{\vec{r}}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

➔ you get 2 sets of **differential equations** you need to solve for r_x, r_y, r_z and v_x, v_y, v_z

Propagation of unperturbed orbits: Newton method

- Keplerian orbit defined by 6 Kepler elements

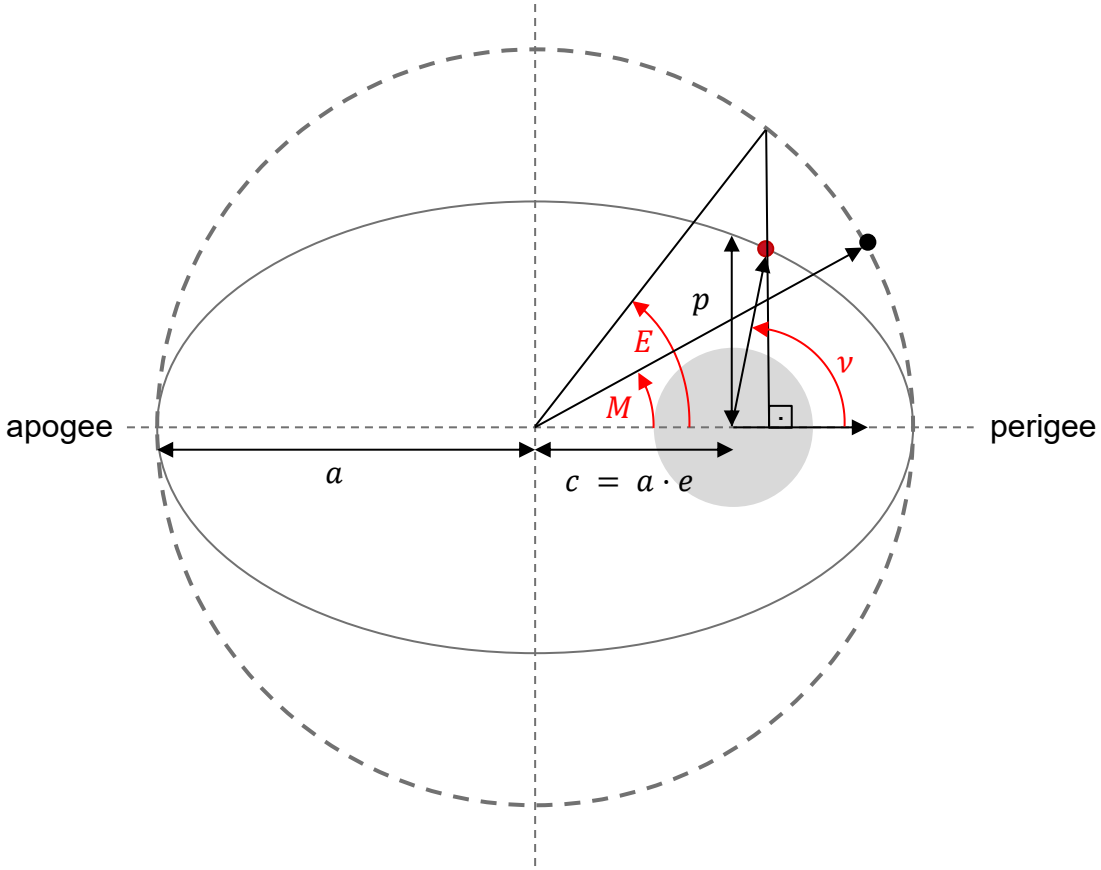
  ➔ “ideal orbit”: no perturbations are taken into consideration

- Kepler equation:

$$M = E(t) - e \sin E(t) = M_0 + n(t - t_0)$$

- Newton method for a numerical solution of E
- Analytical solution of true anomaly:

$$\cos E = \frac{e + \cos \nu}{1 + e \cos \nu} \qquad \cos \nu = \frac{\cos E - e}{1 - e \cos E}$$

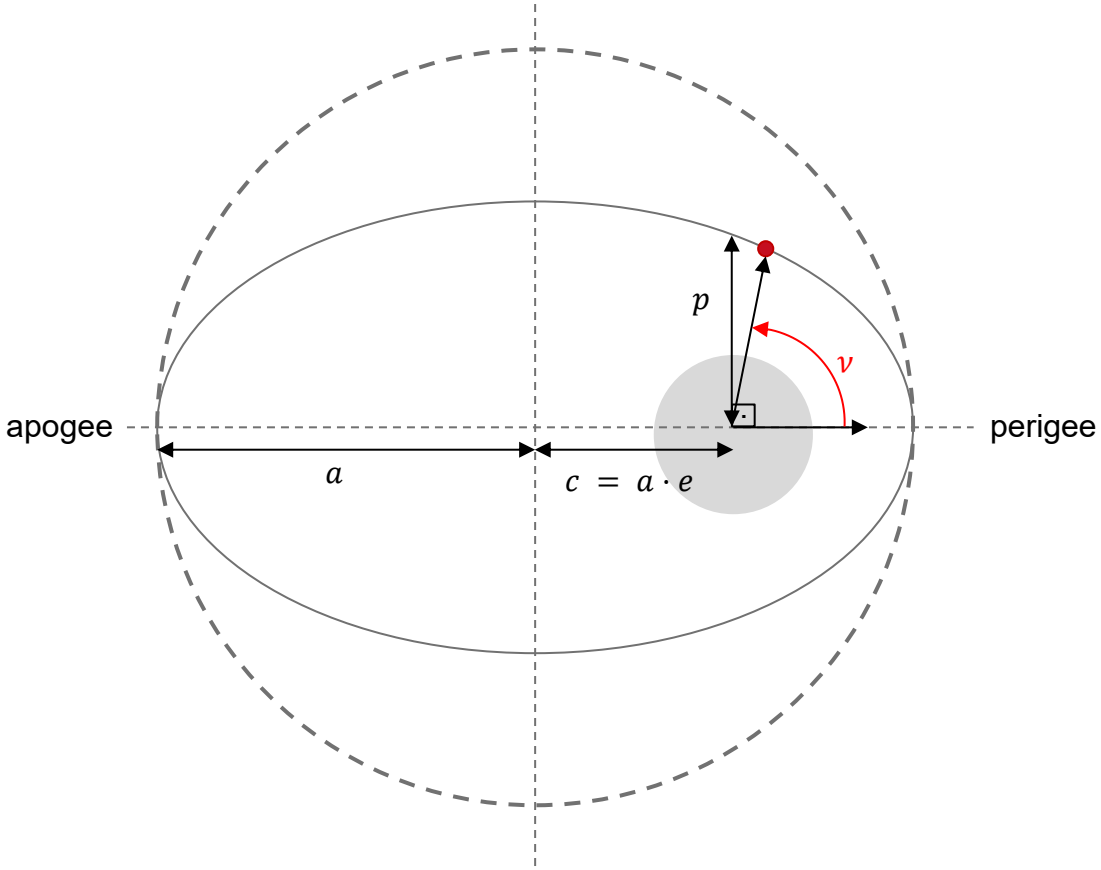


Recap: True Anomaly ν

… geocentric angle between perigee and true position of the satellite in its orbit plane

- Given by parameters of the ellipse:

$$r = \frac{a(1-e^2)}{1+e\,cos\nu}$$



Recap: Eccentric Anomaly E

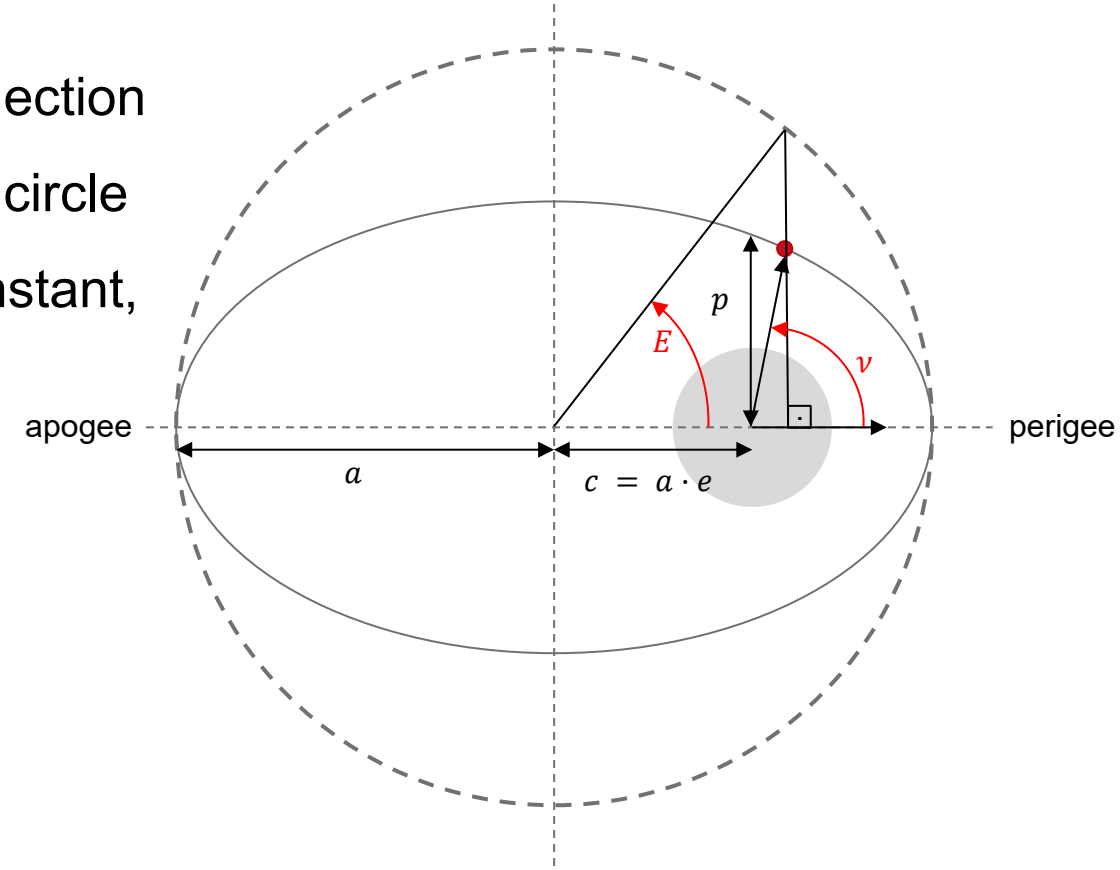
…angle between the perigee and the vertical projection of the position of the satellite onto the auxiliary circle

- Depends on semi-major axis, gravitational constant, mean motion
- Kepler's equation:

$$E(t) - e \sin E(t) = n(t - t_0)$$

- Relation with true anomaly:

$$\cos E = \frac{e + \cos \nu}{1 + e \cos \nu} \qquad \cos \nu = \frac{\cos E - e}{1 - e \cos E}$$

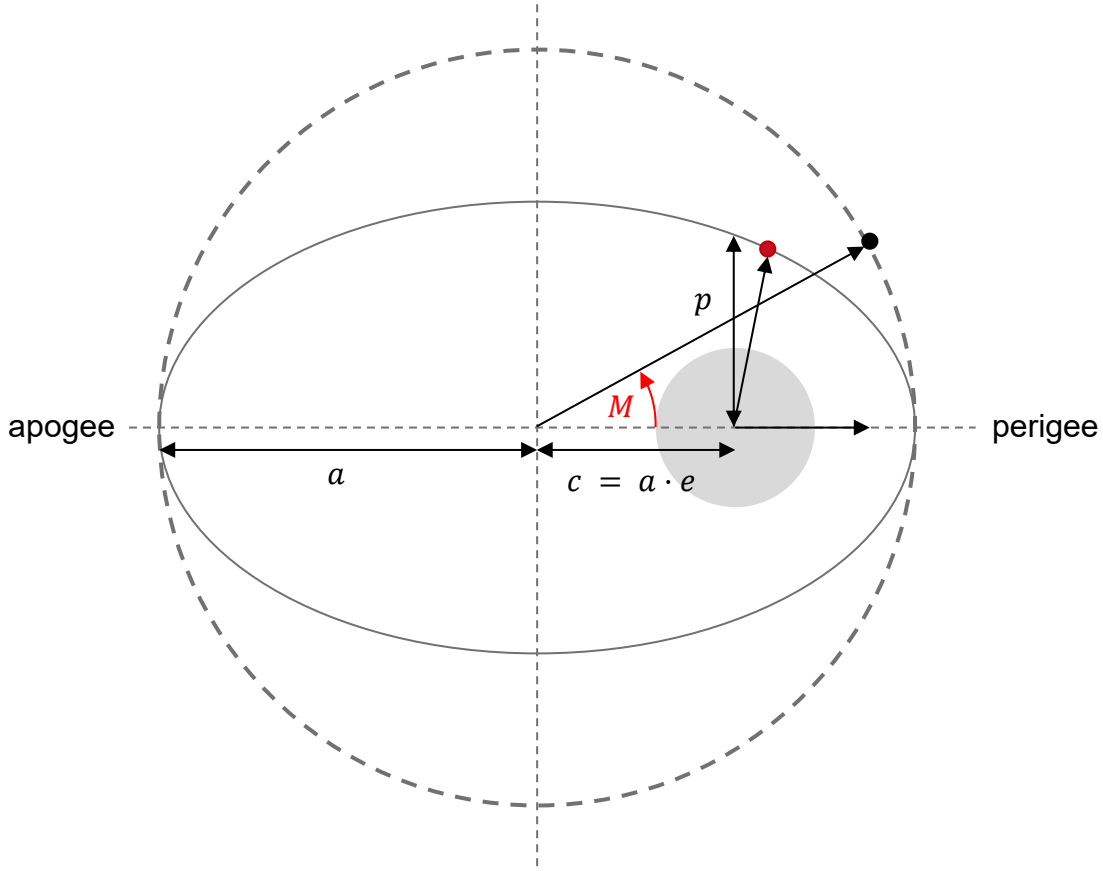


Recap: Mean Anomaly M

… angle between the perigee and a point on a circular orbit where the satellite would have been circling with the mean motion n

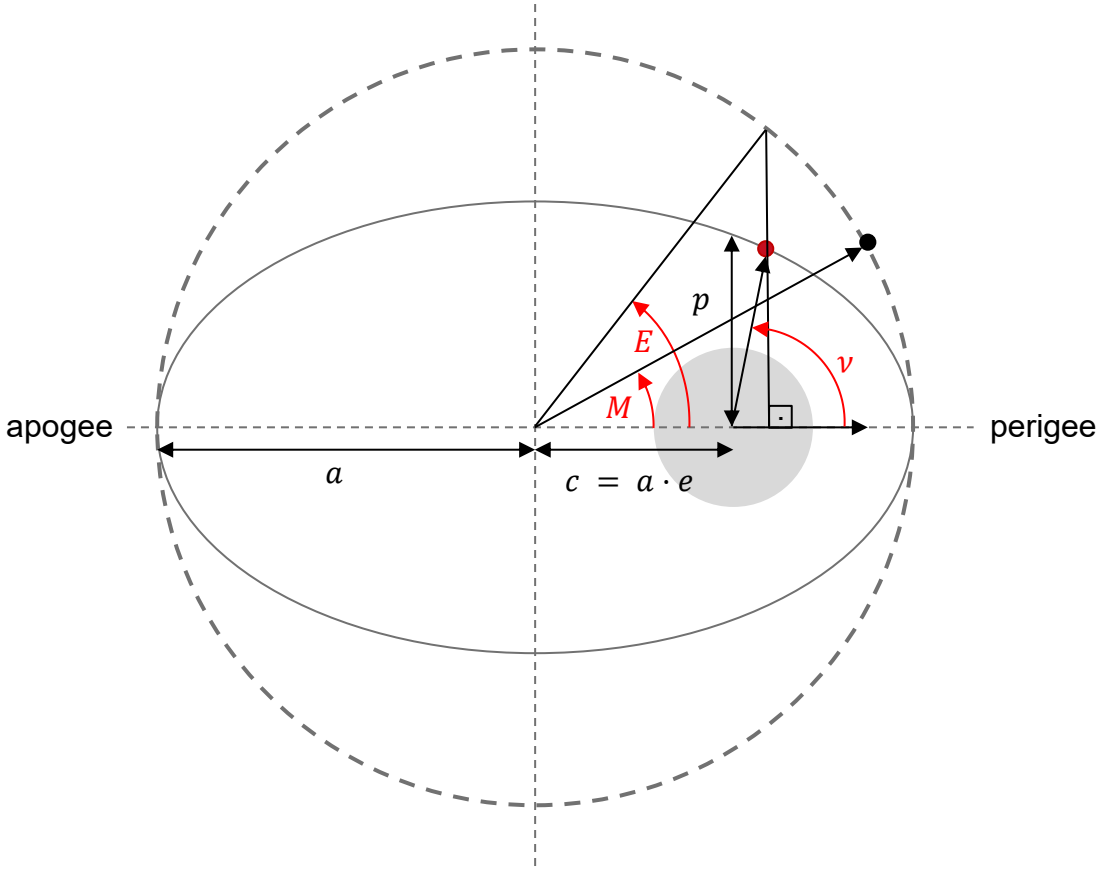
- Kepler's equation

$$M = E(t) - e \sin E(t) = M_0 + n(t - t_0)$$



Propagation (1)

- Goal: Find position of satellite at t
- Given: state of the satellite as Keplerian elements a_0, e_0, i_0, Ω_0, ω_0, ν_0 at t_0
  - For unperturbed orbits only ν changes with time
- Propagate the satellite's state using the Kepler equation
  - Mean anomaly M changes linearly with time



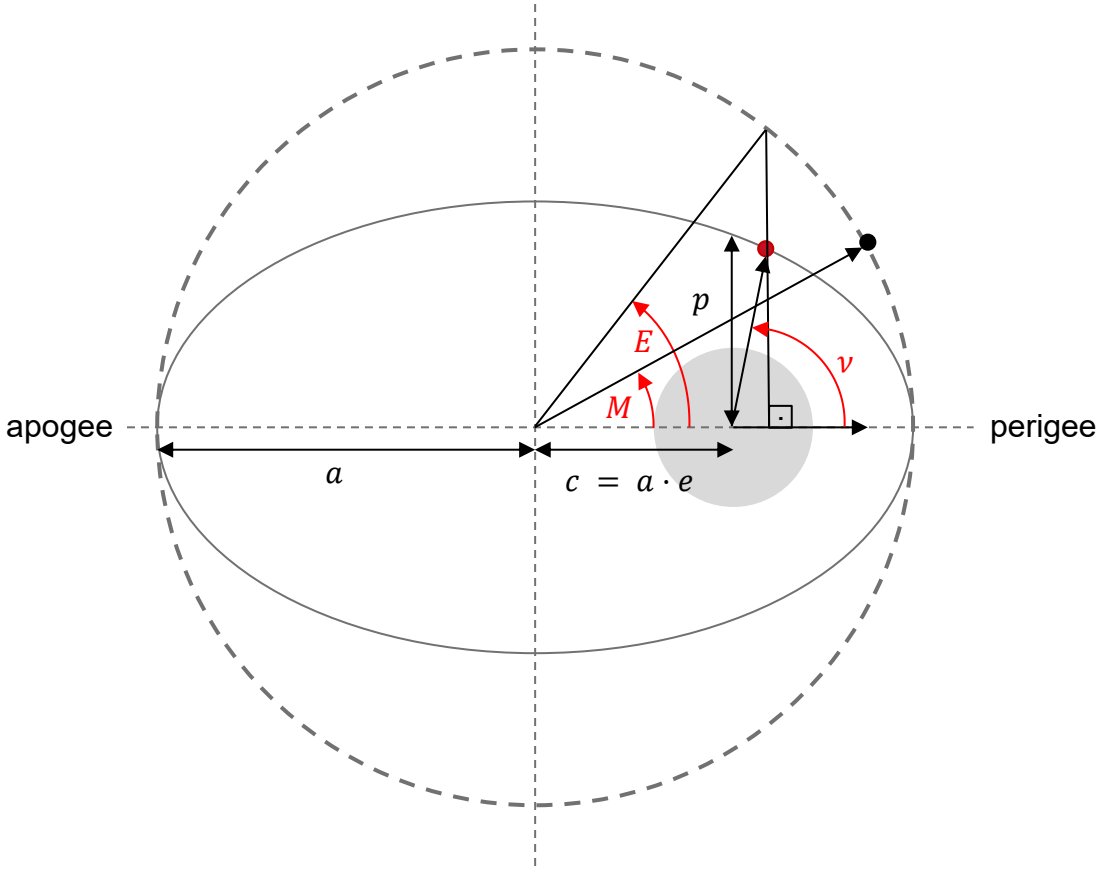
Propagation (2)

- Change of M is calculated using the mean motion n:

$$n = \frac{2\pi}{P} = \sqrt{\frac{\mu}{a^3}}$$

- n is the mean angular speed of the satellite
- Integrating n from t_0 to t yields:

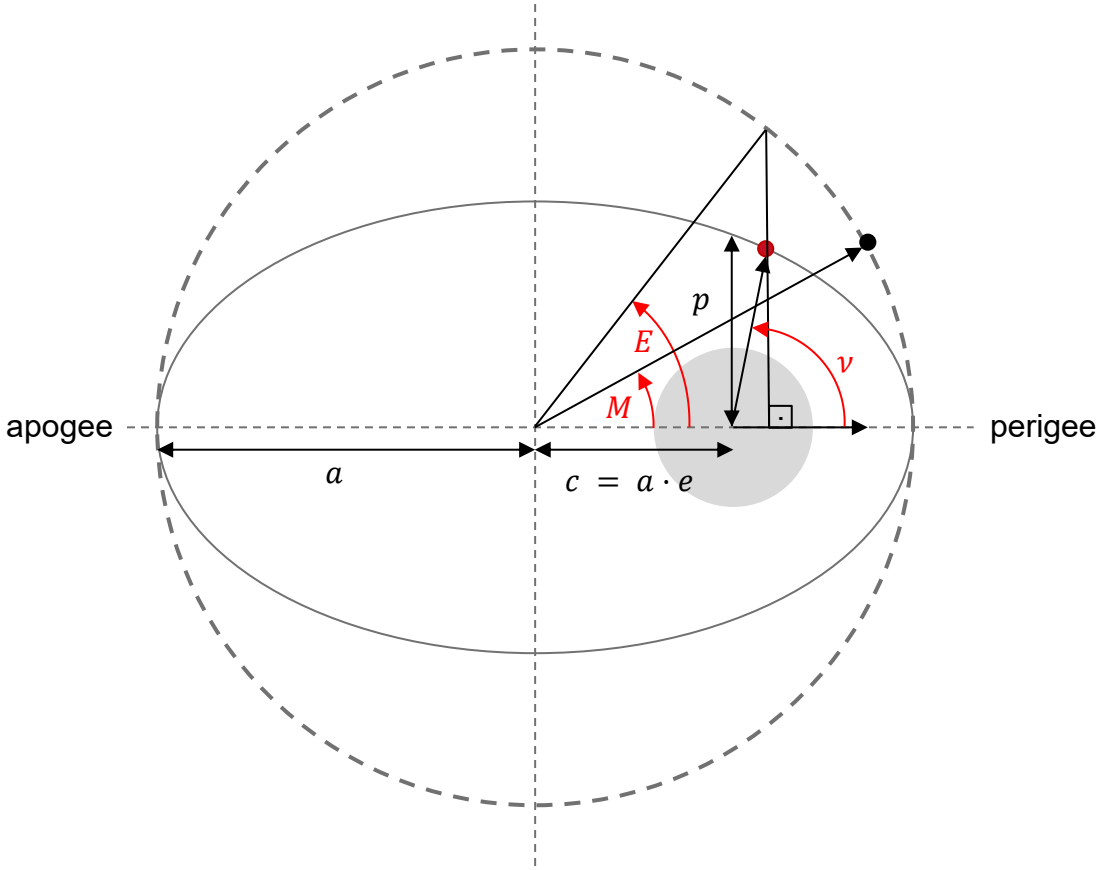
$$\Delta M = n(t - t_0)$$



Propagation (3)

- To propagate the state of the satellite, i.e. computing the state at t with the given initial state at t_0, ΔM is needed
- M_0 is calculated with ν_0 via E_0
- Kepler‘s equation is not analytically solvable

$$M = E(t) - e \sin E(t) = M_0 + n(t - t_0)$$



Solution of Kepler's Equation (1)

- Various methods to solve the equation
- An easy and fast way: Newton method

  ➔ basis: $v = f(t)$ and $f(E) = E - e\sin(E) - M$

$$E_{i+1} = E_i - \frac{f(E_i)}{f'(E_i)} = E_i - \frac{E_i - e\sin E_i - M}{1 - e\cos E_i}$$

- Newton method and solution of Kepler's equation will converge
- For small e: $E_0 = M$, for $e > 0.8$: $E_0 = \pi$
- Index 0 here is not initial state, but initial value for Newton method

Solution of Kepler's Equation (2)

- With E obtained with Newton method, state at t is calculated
- Only ν changes compared to initial state t_0

  ➔ Due to perturbation-free orbit

Exercise/Homework Prep 1

- Implement the transformation of Keplerian elements to cartesian coordinates as a function in MATLAB

Where to start:

- Think of inputs and outputs (Vectors? Matrices?)
- What should be parametrized and what can be hard-coded?
- What do you need for the transformation? Try to visualize the transformation on paper.
- For testing: visualize an orbit with $\Omega = 0$, $i = 0$, $\omega = 0$, $\nu = 0$. Where is the satellite?

Transformation Keplerian ←→ Cartesian

1. Step: express orbit height and semi-latus rectum as a function of Keplerian elements

$$r = \frac{a(1-e^2)}{1+e\,cos\nu} \qquad p = a(1-e^2)$$

1. Step: express r in the perifocal COS $\quad \boldsymbol{r}_{perif.} = \begin{pmatrix} \cos\nu \\ \sin\nu \\ 0 \end{pmatrix} r \qquad \boldsymbol{v}_{perif.} = \begin{pmatrix} -\sin\nu \\ e+\cos\nu \\ 0 \end{pmatrix} \sqrt{\frac{\mu}{p}}$

2. Step: define rotation matrices:

$$\boldsymbol{R_\Omega} = \begin{pmatrix} \cos\Omega & \sin\Omega & 0 \\ -\sin\Omega & \cos\Omega & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \boldsymbol{R_i} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos i & \sin i \\ 0 & -\sin i & \cos i \end{pmatrix} \quad \boldsymbol{R_\omega} = \begin{pmatrix} \cos\omega & \sin\omega & 0 \\ -\sin\omega & \cos\omega & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

➔ $\boldsymbol{r}_{cart} = \boldsymbol{R_\Omega R_i R_\omega r}_{perif.} \;\; \boldsymbol{v}_{cart.} = \boldsymbol{R_\Omega R_i R_\omega v}_{perif.}$

Exercise/Homework Prep 2

- Implement the Newton method as a function in MATLAB

Where to start:

- Think of inputs and outputs
- What should be parametrized and what can be hard-coded?
- Consider break conditions

Perturbations

Perturbations

- Various perturbations act on a satellite:
  - Conservative forces: gravitation

    → total energy constant

    → Can be modelled as gradient of scalar potential function
  - Non-conservative forces: solar radiation, drag, thrust

    → total energy changes
- Modelling them helps better approximate true motion of a satellite for accurate mission analysis

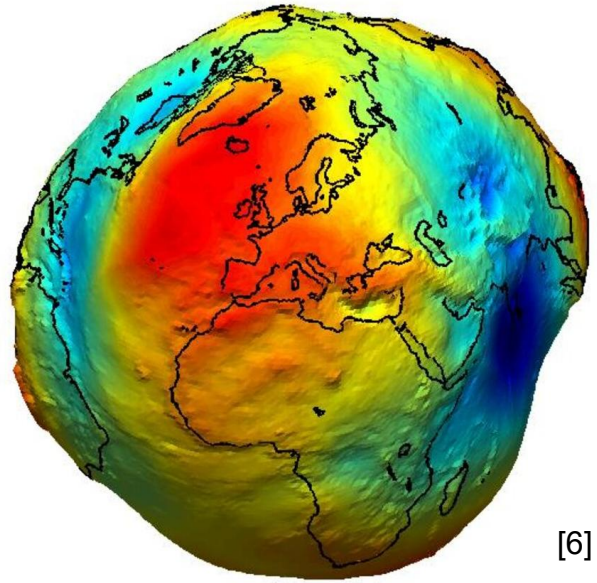
Gravitational Potential

- Earth's gravitational force is expressed as an acceleration
- Modelled with a potential

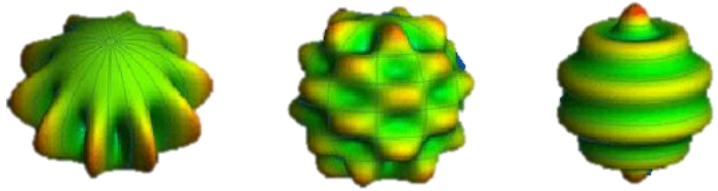
$$U(r,\phi,\lambda) = \frac{\mu}{r}\sum_{n=0}^{\infty}\sum_{m=0}^{n}\frac{R^n}{r^n}\cdot P_{nm}(sin\,\phi)\cdot(C_{nm}\cdot cos(m\lambda)+S_{nm}\cdot sin(m\lambda))$$

with P_{nm} being a Legendre polynomial

- A simplified model can be given considering only zonal harmonics (J2, J3, etc.)
- $J_n = -C_{n0} \qquad J_{nm} = -C_{nm}$
- Coefficients also exists as normalized coefficients, e.g. $\bar{C}_{nm}$



[6]



Sectorial Tesseral Zonal

Third Body Perturbations

- Gravity of other bodies of the Solar System influences satellite
- Main influences at Earth are Moon and Sun

$$\ddot{\boldsymbol{r}}_B = \mu_B \left(\frac{\boldsymbol{r_B} - \boldsymbol{r}}{|\boldsymbol{r_B} - \boldsymbol{r}|^3} - \frac{\boldsymbol{r_B}}{r_B^3} \right)$$

with

- Position vector of body $\boldsymbol{r_B}$
- Gravitational parameter of body μ_B

Atmospheric Drag

- Atmosphere of Earth causes drag
- Retrograde perturbation, i.e. in negative velocity direction

$$\ddot{\boldsymbol{r}}_{Drag} = -\frac{1}{2} C_D \frac{A}{m} \rho v^2 \frac{\boldsymbol{v}}{|v|}$$

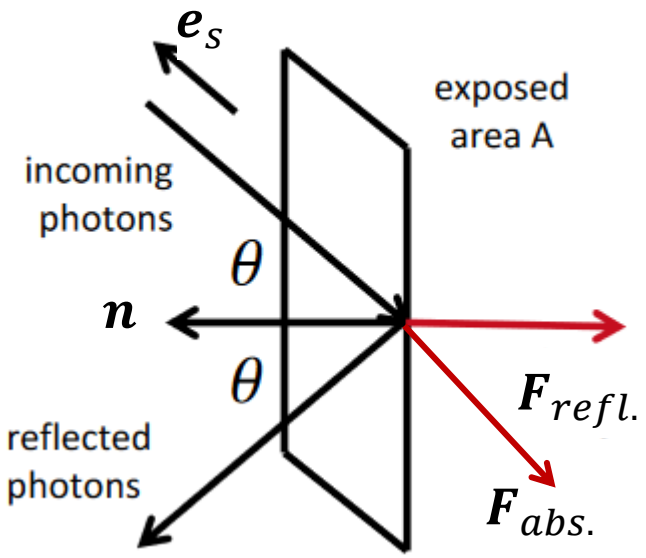
with

- Density of atmosphere at satellite position ρ
- Drag coefficient C_D (dimensionless parameter, typical values 1.5-3. Depends on the shape)

Solar Radiation Pressure (1)

- Incident photons carry momentum
- Momentum is transferred to satellite when photons are absorbed and reflected
- Pressure P_S is dependent on solar flux Φ [W/m²]
  - Solar flux changes proportional to $\frac{1}{{r_s}^2}$ (inverse square law with distance from Sun r_s)

$$P_S = \frac{\Phi}{c}\,, \quad \text{with speed of light } c$$



Solar Radiation Pressure (2)

Total absorption

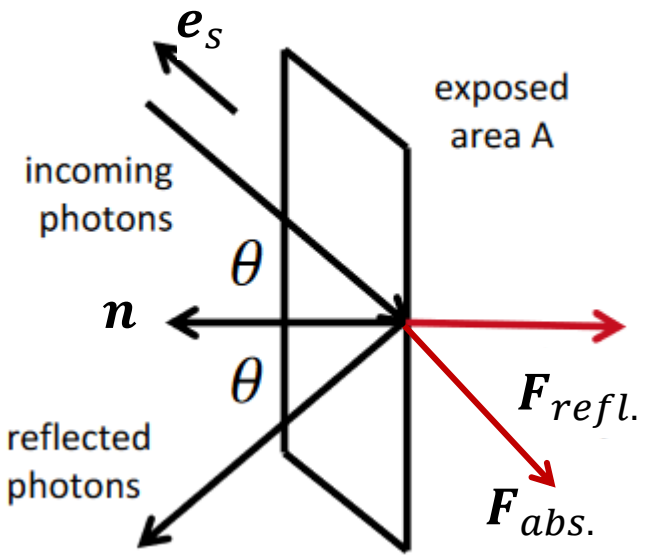
$$\boldsymbol{F}_{abs} = -P_s A \cos\theta \, \boldsymbol{e}_s$$

Total reflection

$$\boldsymbol{F}_{refl.} = -2P_s A \cos\theta \ \ cos\theta \, \boldsymbol{n}$$

Absorption and reflection

$$\boldsymbol{F}_{abs} + \boldsymbol{F}_{refl.}$$



Solar Radiation Pressure (3)

- Simplifications: cross-sectional area A, satellite in orbit around Earth
  - Assuming surface normal $\boldsymbol{n}$ points towards Sun

$$\ddot{\boldsymbol{r}}_{SRP} = -P_S C_R \frac{A}{m} \frac{\boldsymbol{r_S}}{r_S^3} AU^2$$

with

– Radiation pressure coefficient $C_R = 1 + \epsilon$ (dimensionless)
  - Reflectivity of surface ϵ

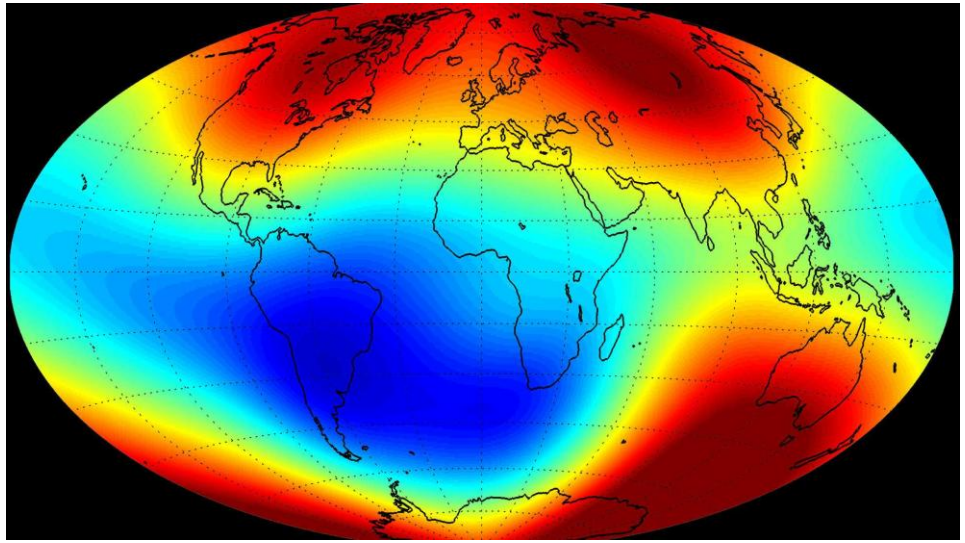
– Radiation pressure (at Earth) $P_S \approx 4.56 \cdot 10^{-6}$ N/m²

Geomagnetic Force

- A standard model used for e.g. propagation is IGRF

  → geomagnetic field is modelled as a gradient of a scalar potential

  → Geomagnetic potential model can be then expressed as:

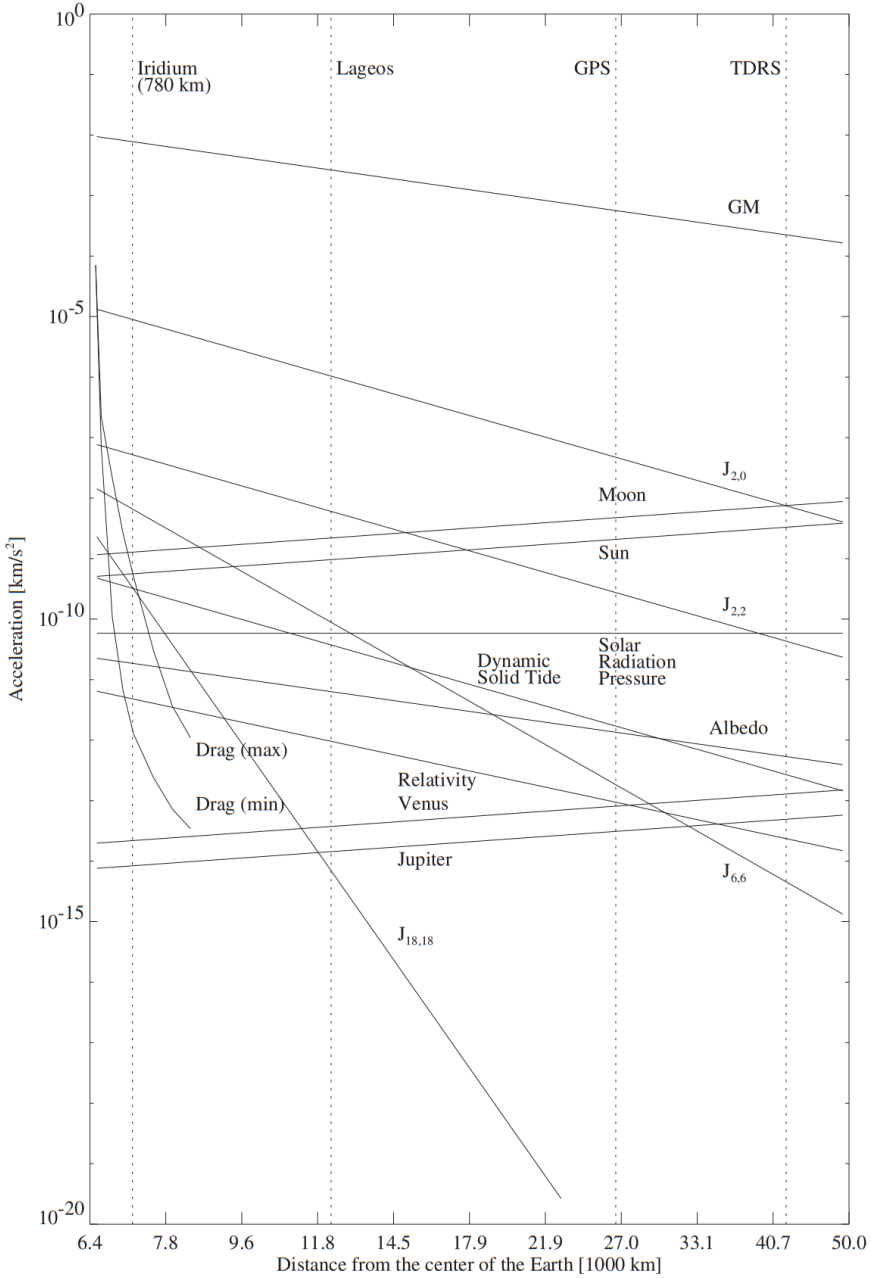


[7]

$$U(r,\varphi,\lambda,t) = R_E \left(\sum_{n=1}^{N} \sum_{m=0}^{n} \left(\frac{R_E}{r}\right)^{n+1} [g_n^m(t)\cos(m\lambda) + h_n^m(t)\sin(m\lambda)] P_n^m(\sin\varphi) \right)$$

Magnitude of Perturbations

- Assess magnitude to decide which perturbations to model
- At Earth, J2 dominates, other coefficents cause smaller perturbations
- Gravitational parameter GM, i.e μ, expresses gravity in this figure
- Drag differs between minimum and maximum depending on the solar cycle



[8]

Modelling Perturbed Orbits

Equation of Motion

- General differential equation:

$$\ddot{\boldsymbol{x}} = \boldsymbol{f}(t, \boldsymbol{x}) \quad \leftrightarrow \quad \boldsymbol{x}(t) = \boldsymbol{x}(t_0) + \int_{t_0}^{t} \boldsymbol{f}\big(t, \boldsymbol{x}(t)\big) dt$$

- Equation of motion:

$$\ddot{\boldsymbol{r}} = \boldsymbol{f}(t, \boldsymbol{r}) = -\frac{\mu}{r^3}\boldsymbol{r}$$

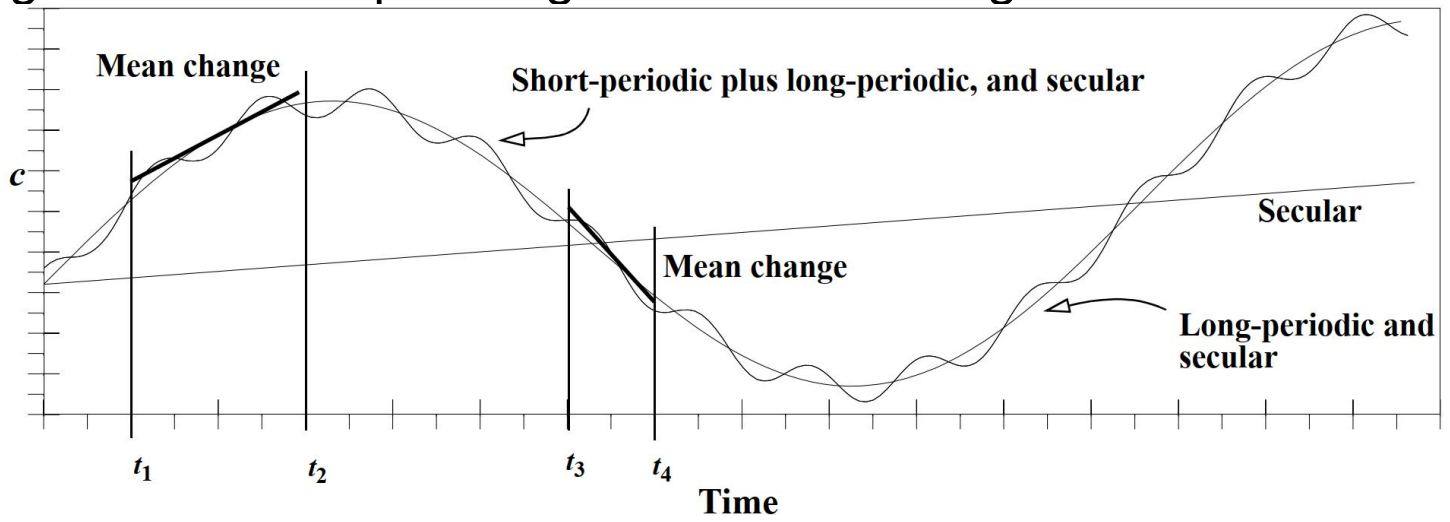
- Equation of motion with perturbations:

$$\ddot{\boldsymbol{r}} = \boldsymbol{f}(t, \boldsymbol{r}) = -\frac{\mu}{r^3}\boldsymbol{r} + \underbrace{\boldsymbol{a}_{grav.}(r,t) + \boldsymbol{a}_{SRP}(r,t) + \boldsymbol{a}_{drag}(r,t) + \boldsymbol{a}_{thrust}(r,t)}_{\boldsymbol{a}_{pertub.}(r,t)}$$

Mean Orbital Elements

… orbital elements averaged over time period to eliminate short- (and long-) periodic fluctuations

- Offer fast approximations of satellite's motion
- Used for long-term mission planning and mission design



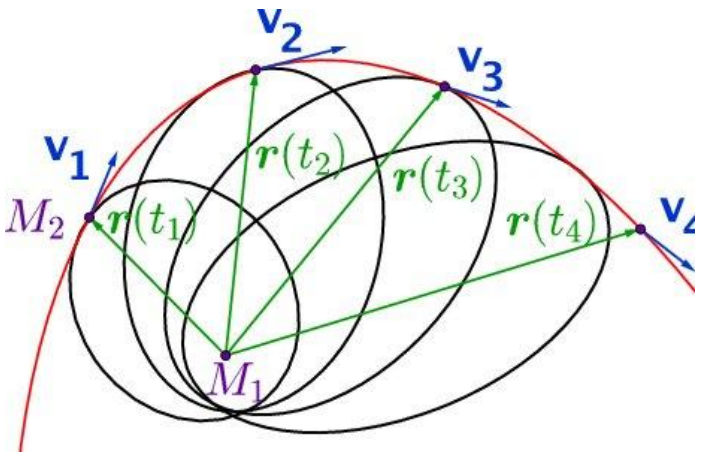
[9]

Osculating Orbital Elements

… true time-varying orbital elements

- Include long and short periodic effects
- Used for high-precision simulations, real-time pointing and tracking operations
- Each set of osculating elements form an ellipse

  ➔ osculating ellipse: fictional two-body orbit if no perturbations at that time step



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Solving Equation of Motion With Osculating Elements

General perturbation techniques

- Analytical solution for any initial conditions
- Limited accuracy due to generalist approach
- Used mainly for mission design, studies and long-term predictions

Special perturbation techniques

- Numerical solution for a set of initial input parameters/conditions
- High precision but high computational burden
- Used for high-precision orbit simulations and determination

General Perturbation Techniques

General Perturbation Techniques

- Equation of motion in cartesian coordinates with perturbations is analytically not solvable

  ➔ use osculating elements instead
- rewrite equation of motion using osculating elements (method of variation of parameters)

  ➔ analytical approximation using a series expansion of the perturbations

  ➔ output: orbital elements for a certain time interval
- Advantage: qualitative behavior and trends can be analyzed
- Dependent on orbit type but not on initial conditions

General Perturbation Techniques

- Two-body problem without perturbations:

$$\ddot{\boldsymbol{r}} + \frac{\mu}{r^3}\boldsymbol{r} = \boldsymbol{0}$$

$$\boldsymbol{r} = \boldsymbol{f}(t, a, e, i, \Omega, \omega, M)$$

$$\boldsymbol{r} = \boldsymbol{f}(t, c_1, c_2, c_3, c_4, c_5, c_6) \; with \; c_i = const. \; i = 1{:}6$$

- Now with perturbations (arbitrary force F) and considering osculating elements

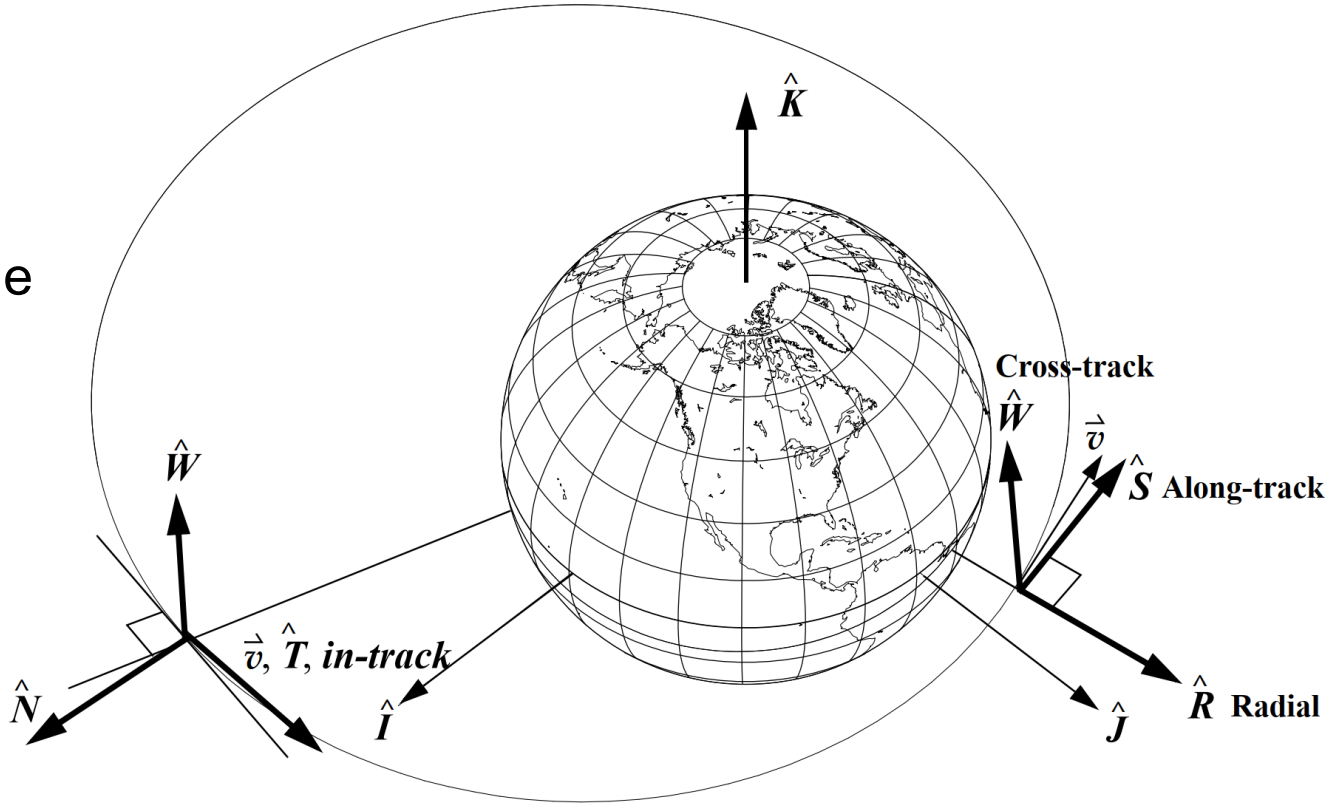
$$\ddot{\boldsymbol{r}} + \frac{\mu}{r^3}\boldsymbol{r} = \boldsymbol{F}$$

$$\boldsymbol{r} = \boldsymbol{f}(t, c_1, c_2, c_3, c_4, c_5, c_6) \; with \; c_i = c_i(t), \; i = 1{:}6$$

➔ 2 main approaches to express: Lagrange and Gaussian

Definition of COS for Perturbation Forces

- Forces and accelerations are given in cartesian coordinates
- Usually RSW system, originating in the satellite
  - RSW is radius-vector based



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Lagrange Method

- Offer a solution for conservative perturbation forces only
- Assumption: all conservative forces can be summed up by a potential function ***R(r)***
- Equation of motion becomes:

$$\ddot{\boldsymbol{r}} + \frac{\mu}{r^3}\boldsymbol{r} = \frac{\boldsymbol{\partial R}}{\boldsymbol{\partial r_i}}$$

- Using the partial derivatives of R on the six Keplerian elements one gets the Lagrange's Planetary Equations

Lagrange's Planetary Equations

$$\frac{da}{dt} = \frac{2}{na}\frac{\partial R}{\partial M_0}$$

$$\frac{de}{dt} = \frac{1-e^2}{ena^2}\frac{\partial R}{\partial M_0} - \frac{\sqrt{1-e^2}}{ena^2}\frac{\partial R}{\partial \omega}$$

$$\frac{di}{dt} = \frac{1}{na^2\sqrt{1-e^2}\sin i}\left(\cos i\frac{\partial R}{\partial \omega} - \frac{\partial R}{\partial \Omega}\right)$$

$$\frac{d\omega}{dt} = \frac{\sqrt{1-e^2}}{ena^2}\frac{\partial R}{\partial \mathrm{e}} - \frac{\cot i}{na^2\sqrt{1-e^2}}\frac{\partial R}{\partial \mathrm{i}}$$

$$\frac{d\Omega}{dt} = \frac{1}{na^2\sqrt{1-e^2}\sin i}\frac{\partial R}{\partial i}$$

$$\frac{dM_0}{dt} = -\frac{1-e^2}{ena^2}\frac{\partial R}{\partial e} - \frac{2}{na}\frac{\partial R}{\partial a}$$

Application of Lagrange Method (1)

- Orbit propagation for asteroids and comets
- Investigations of Earth's gravitation only
- **Example: effect of the oblateness of Earth (only J2 term):**

$$R = \frac{\mu}{a^3}\frac{a^3}{r^3}\left(\frac{2\sin^2 i - 2}{4} - \frac{3}{4}\sin^2 i \cos 2(v+\omega)\right)$$

→ let's consider only the secular changes (for simplicity)

→ using Fourier series expansion of ***R*** and the elimination of periodic terms one gets

$$R_0 = \frac{\mu}{a^3}\left(\frac{3\sin^2 i - 2}{4}(1-e^2)^{-\frac{3}{2}}\right)$$

Application of Lagrange Method (2)

Example: effect of the oblateness of Earth (only J2 term):

$$R_0 = \frac{\mu}{a^3}\left(\frac{3\sin^2 i - 2}{4}(1-e^2)^{-\frac{3}{2}}\right)$$

➔ R_0 only dependent on a, i and e ➔ simplifies the differential equations:

$$\frac{da}{dt} = \frac{2}{na}\frac{\partial R}{\partial M_0}$$

$$\frac{de}{dt} = \frac{1-e^2}{ena^2}\frac{\partial R}{\partial M_0} - \frac{\sqrt{1-e^2}}{ena^2}\frac{\partial R}{\partial \omega}$$

$$\frac{di}{dt} = \frac{1}{na^2\sqrt{1-e^2}\sin i}\left(\cos i\frac{\partial R}{\partial \omega} - \frac{\partial R}{\partial \Omega}\right)$$

$$\frac{d\omega}{dt} = \frac{\sqrt{1-e^2}}{ena^2}\frac{\partial R}{\partial \mathrm{e}} - \frac{\cot i}{na^2\sqrt{1-e^2}}\frac{\partial R}{\partial \mathrm{i}}$$

$$\frac{d\Omega}{dt} = \frac{1}{na^2\sqrt{1-e^2}\sin i}\frac{\partial R}{\partial i}$$

$$\frac{dM_0}{dt} = -\frac{1-e^2}{ena^2}\frac{\partial R}{\partial e} - \frac{2}{na}\frac{\partial R}{\partial a}$$

Gaussian Method

- Offers a solution for both conservative and non-conservative forces
- Approach based on the consideration of the specific forces

$$\boldsymbol{a}_{pert.} = (R, S, W)$$

$$R = \frac{\boldsymbol{r}}{r}\,\boldsymbol{F}, \qquad S = \frac{(\boldsymbol{r} \times \dot{\boldsymbol{r}}) \times \boldsymbol{r}}{|\boldsymbol{r} \times \dot{\boldsymbol{r}}| r}\,\boldsymbol{F}, \qquad W = \frac{(\boldsymbol{r} \times \dot{\boldsymbol{r}})}{|\boldsymbol{r} \times \dot{\boldsymbol{r}}|}\,\boldsymbol{F}$$

where $\boldsymbol{F}$ is an arbitrary perturbation force and $\boldsymbol{r}$ the position vector

Gaussian Perturbation Equations

$$\frac{da}{dt} = \frac{2a^2}{h}[R \cdot e \sin\nu + S \cdot (1 + \cos\nu)]$$

$$\frac{de}{dt} = \frac{h}{\mu}\left[R \cdot \sin\nu + S \cdot \frac{e + 2\cos\nu + e\cos^2\nu}{1 + e\cos\nu}\right]$$

$$\frac{d\Omega}{dt} = \frac{r}{h}\frac{\sin u}{\sin i} \cdot W, \qquad \frac{di}{dt} = \frac{r}{h}\cos u \cdot W$$

$$\frac{d\omega}{dt} = \frac{h}{\mu}\frac{1}{e}\left[-R\cos\nu + S\frac{2 + e\cos\nu}{1 + e\cos\nu}\sin\nu\right] - \frac{r}{h}\frac{\sin u\cos i}{\sin i}W$$

$$\frac{dM}{dt} = n + \frac{r}{h}\sqrt{1 - e^2}\left[R\left(\cos\nu - \frac{2e}{1 + e\cos\nu}\right) - S\left(1 + \frac{1}{1 + e\cos\nu}\right)\sin\nu\right]$$

Application of Gaussian Method

- Orbit propagation for asteroids and comets
- Trend analyses of single perturbations
- Determination of orbit maneuvers for changes of single parameters

Example: Influence of Atmospheric Drag

- Simple approach: drag creates a force along the track (S)

  ➔ evaluating Gaussian perturbation equations, one sees:

$$\frac{d\Omega}{dt} = 0, \qquad \frac{di}{dt} = 0$$

Application: Simplified General Perturbations Models (SGP)

- First version (SGP) developed in 1960s by U.S. Air Force
- Offers five different orbit propagation models (SGP, SGP4, SDP4, SGP8 and SDP8)
- Two Line Element sets (TLE) are computed with SGP models
- SGP4 has an error of ~1 km at epoch

Image index

[1] https://en.wikipedia.org/wiki/Spherical_trigonometry#/media/File:Spherical_trigonometry_basic_triangle.svg

[2] https://www.gsc-europa.eu/system-service-status/orbital-and-technical-parameters

[3] https://mathworld.wolfram.com/ConicSection.html

[4] Kirkpatrick, D. (1999). *Space mission analysis and design* (Vol. 8). J. R. Wertz, W. J. Larson, & D. Klungle (Eds.). Bloomington, IN: Microcosm.

[5] https://nanopdf.com/download/gladiator-startup-10-5af428eca58f1_pdf

Image index

[6] https://www.esa.int/ESA_Multimedia/Images/2008/05/Earth_s_geoid_as_seen_by_GOCE2

[7] https://earth.esa.int/eogateway/missions/swarm

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[9] Vallado, DA., Wertz, J. *Fundamentals of Astrodynamics and Applications.* 4th ed. New York: Springer; 2013.

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